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Digitalising Social Protection: Better Targeting and Delivery

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BIO

Social assistance in Indonesia misses a large share of the households it is meant for. This asks three questions: how large the gap is, where it comes from, and which parts of it digitalisation can actually close.

THE PROBLEM

We cannot see who is poor. We only see signs.

A hunter cannot see the animal in the tree. He can only see the leaves move. Targeting works the same way: welfare itself is never observed, only proxies for it.

Who decides

Tree A hides the **musang** — the household that qualifies. Tree B holds **birds** — households that look the same from outside.

Exclusion error — the musang escapes: a household that qualifies gets nothing

Inclusion error — a bird is hit: a household that does not qualify receives the benefit

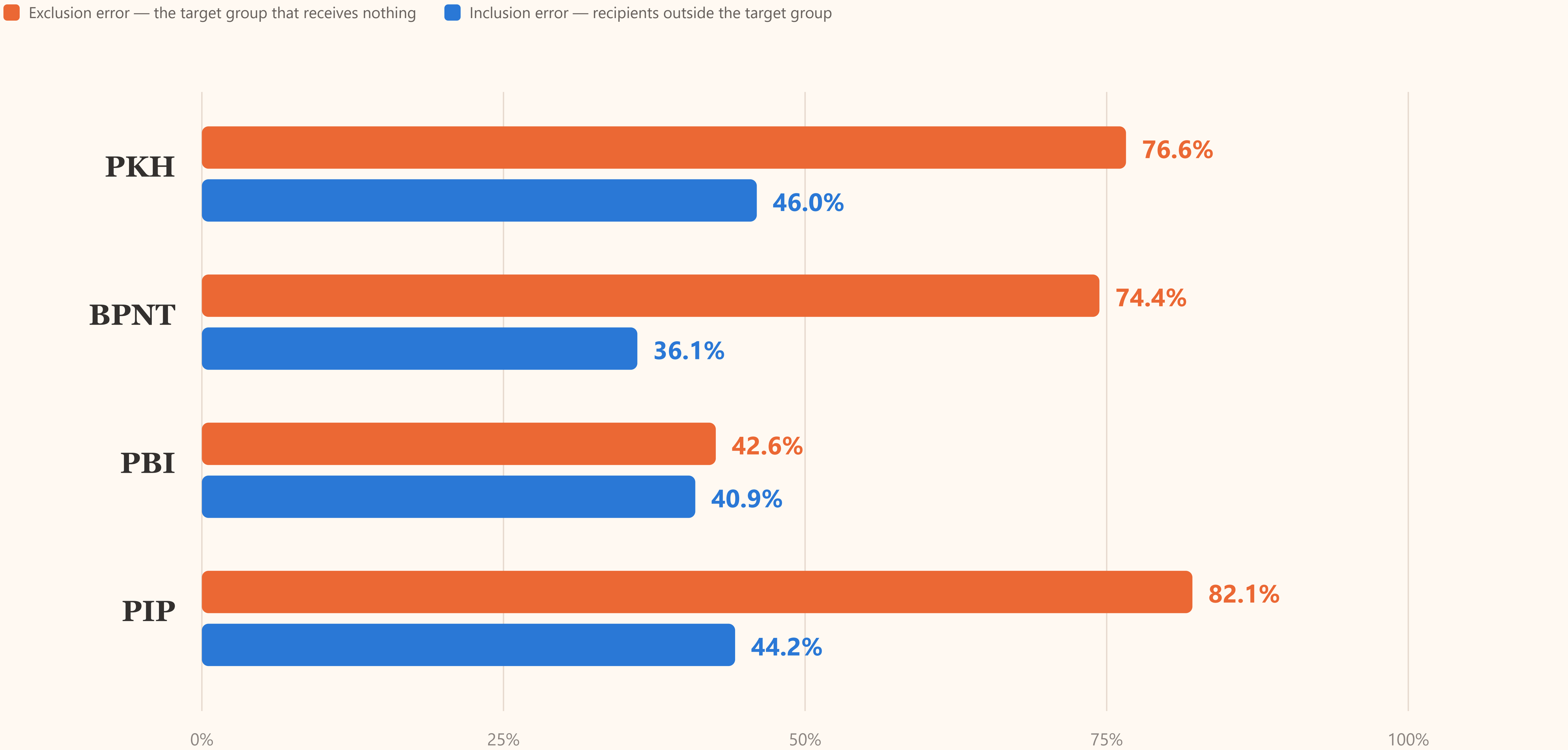
I How well does it reach people?

Four programmes, measured against the households and people they are meant to serve.

FIGURE 1

How much of the target group is missed

Susenas March 2025. Exclusion error is the share of the target group that receives nothing. Inclusion error is the share of recipients who fall outside the target group.



Move the PKH cut-off to deciles 1–3 and inclusion error becomes **57.7%**; move it to 1–5 and it becomes **35.7%**. Nothing about the programme changed — only where the line is drawn.

FIGURE 2

Three years: two programmes improving, one not

Exclusion error, March 2023 to March 2025, on the same definitions. PBI and PIP reach more of their target group each year. PKH does not.

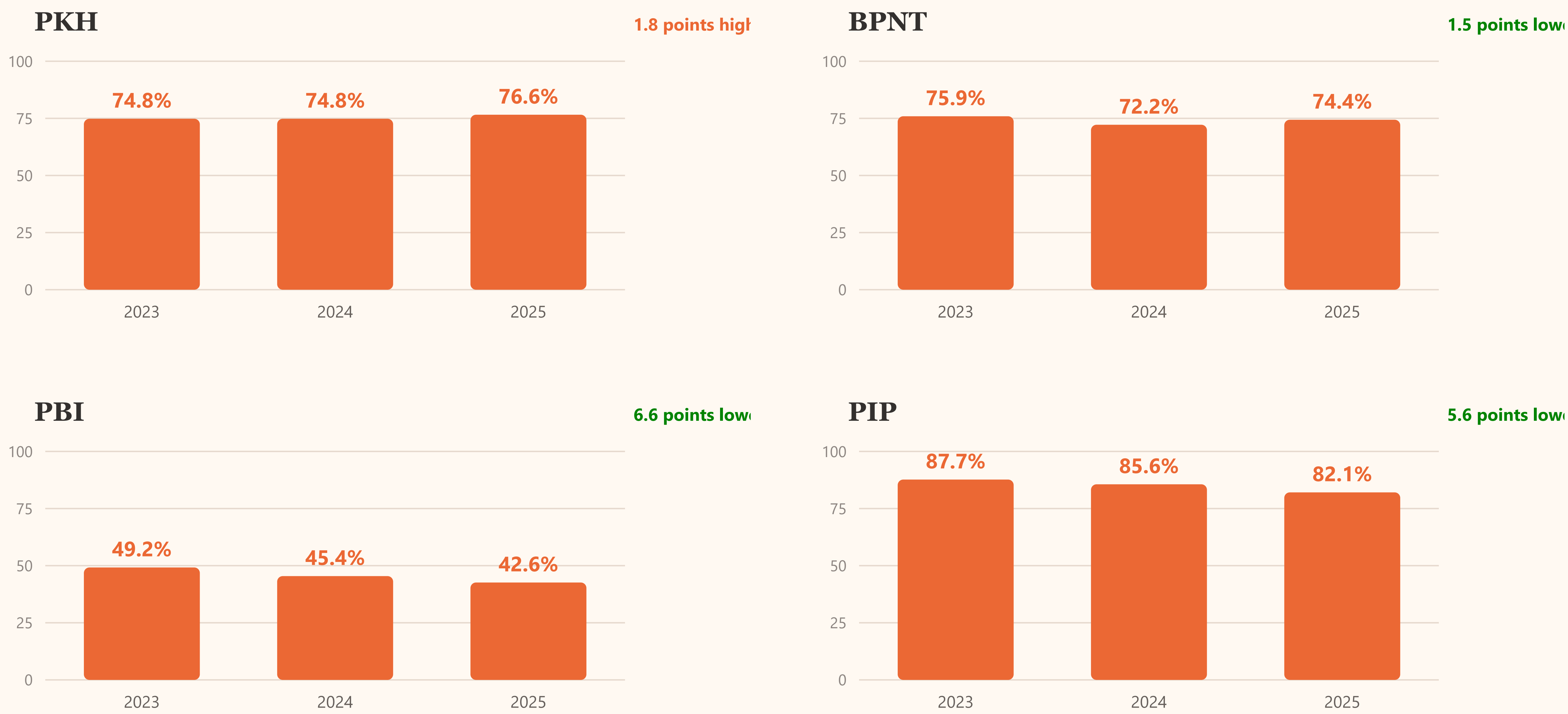


FIGURE 3

The ranking works. The coverage does not.

Share of households or people receiving each programme, by welfare decile. Coverage falls steadily from the poorest decile to the richest in every programme, so the ranking is pointing the right way.

Share receiving the programme

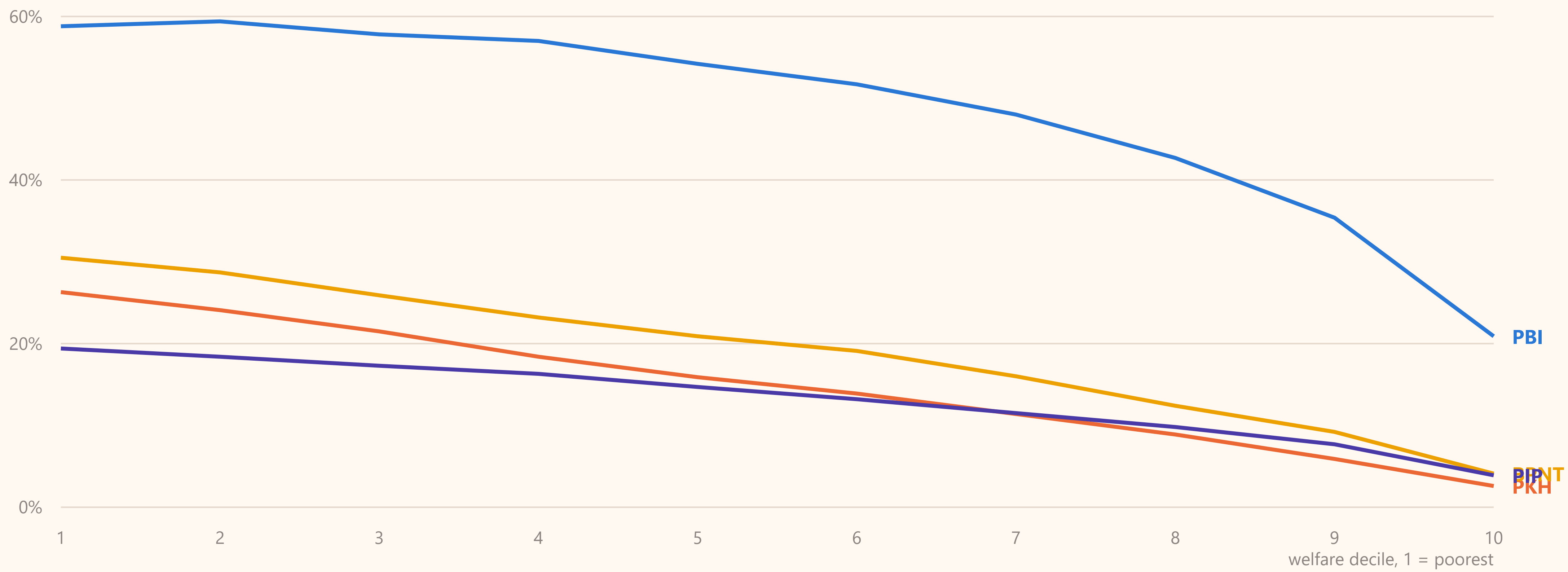
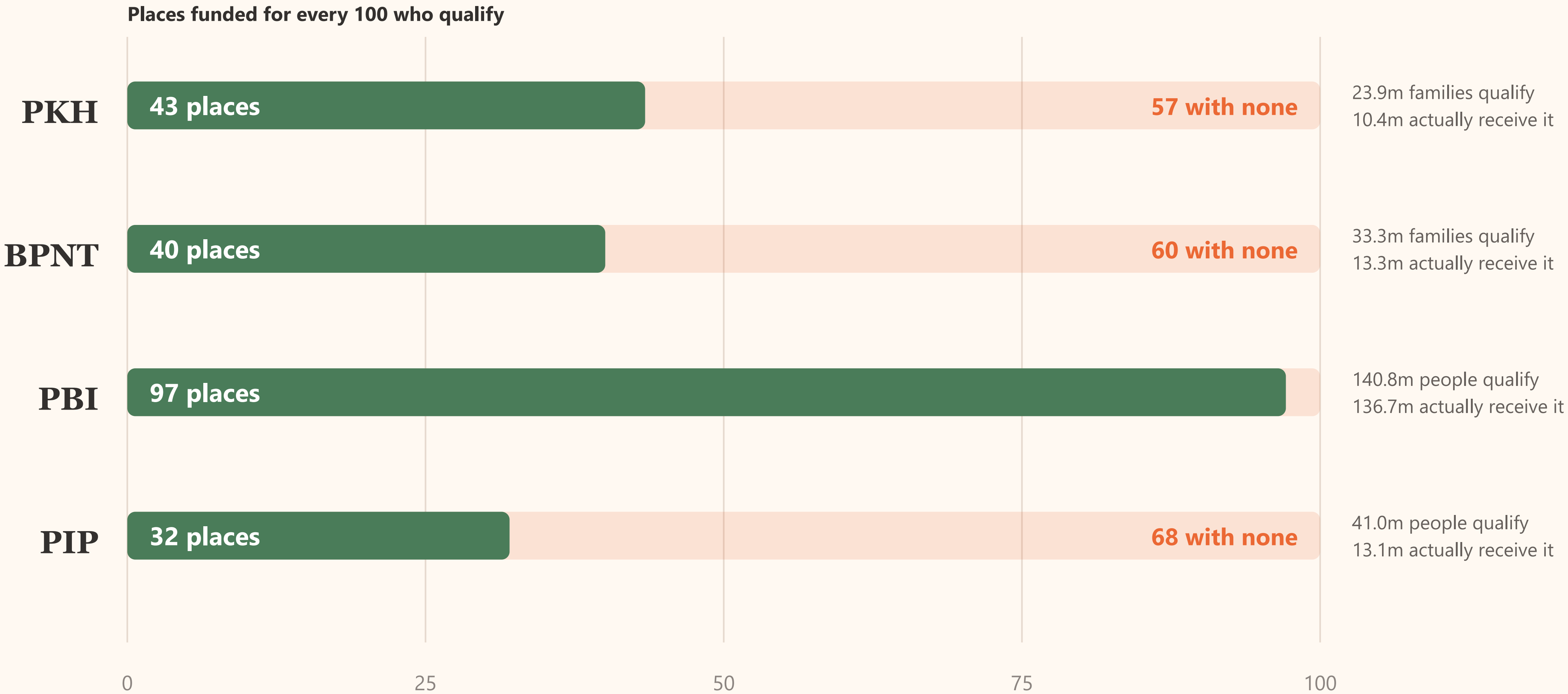


FIGURE 4

There are not enough places for everyone who qualifies

Before asking how well a programme chooses, ask how many places it has. If a programme funds fewer places than there are qualifying families, the rest must be missed no matter how good the targeting is.

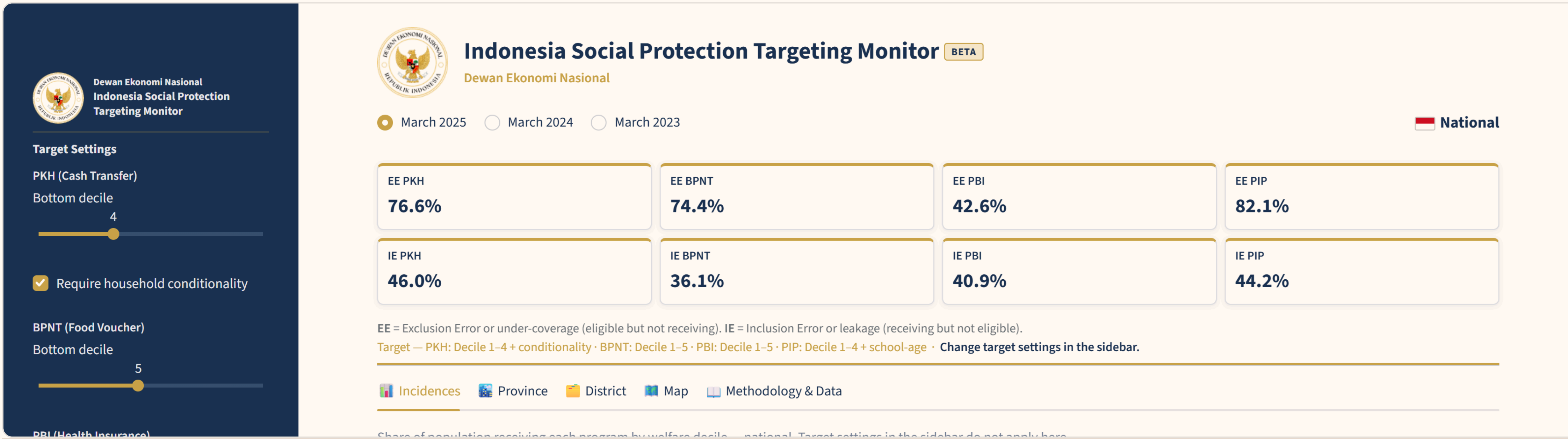


PKH funds 43 places for every 100 families that qualify, so at least 57 of those 100 get nothing whatever the data says. BPNT funds 40, PIP funds 32. **PBI is different**: it funds 97 places for every 100 people who qualify, so almost none of its 42.6% exclusion error can be blamed on the budget — nearly all of it is places going to people outside the target group. That part is what better data fixes.

THE DATA

These numbers are public and can be checked

The Dewan Ekonomi Nasional targeting monitor computes exclusion and inclusion error directly from Susenas microdata, for three years and for every province and district. Anyone can change the target definition and see the numbers move.



II Is this unusual?

Other countries publish the same measures. What their experience shows is where the error actually comes from.

FIGURE 5

The same range everywhere

Exclusion and inclusion error for social assistance programmes with published figures. Exclusion error runs from about 50% to 90%, inclusion error from about 24% to 70%. Indonesia sits inside both ranges.

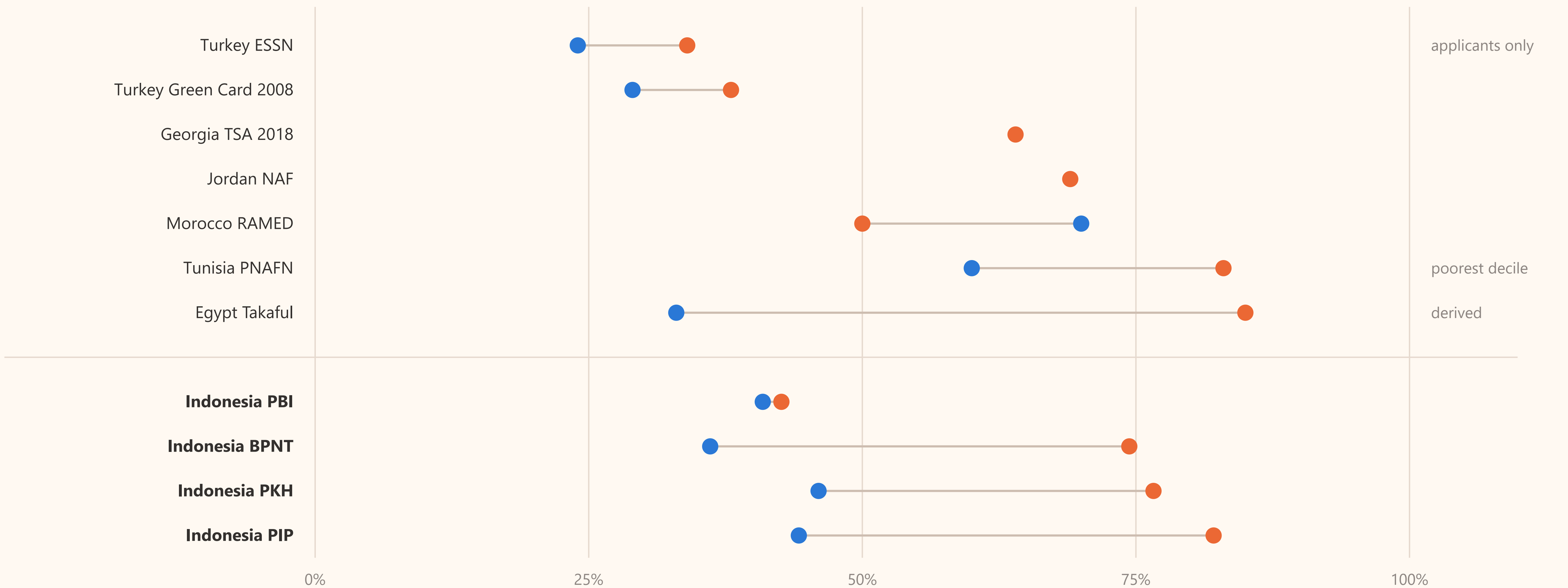
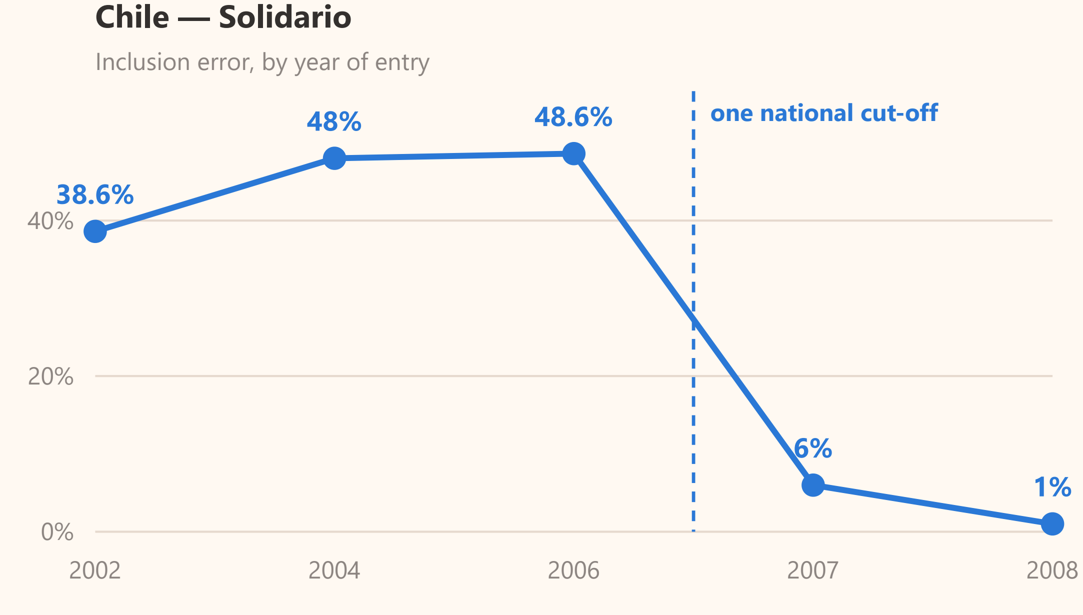
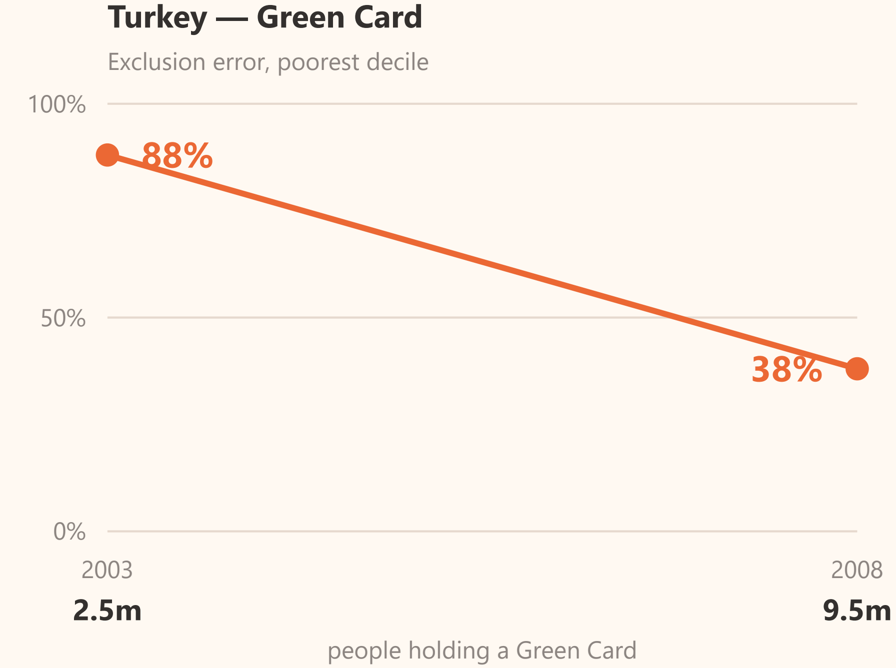


FIGURE 6

What actually changed the numbers elsewhere

Two countries cut their targeting error sharply. Neither did it by improving the statistical model. One changed a rule; the other funded more places.





Chile: a rule, not a formula

Until 2006 each commune had its own cut-off, estimated from a sample of about 200 households. A family could qualify in one commune and be refused across the boundary with the same circumstances.

From 2007 a **single national cut-off** replaced them. Entrants coming from the poorest ventile went from 51% to 99%, and inclusion error fell from 48.6% to 1.0%. **The scoring formula did not change.**

Turkey: more places, not better sorting

Between 2003 and 2008 the Green Card grew from **2.5 million to 9.5 million** holders. Exclusion error among the poorest tenth fell from 88% to 38%.

The expansion was progressive, so the extra places went disproportionately to poor households. The gain came from there being far more places, not from a better method of choosing who filled them.

These are the two levers, and they are the two halves of Figure 4. **Exclusion error falls when there are more places; inclusion error falls when the rule for allocating them improves.** Georgia is the warning in the other direction: its formula was fixed at 2013 prices and never re-indexed, so inflation pushed families above the line without making them better off, and exclusion error rose from 54% to 64%.

III Why is it imperfect?

Two reasons. One is a limit on what any formula can know. The other is everything that happens after the formula.

FIGURE 7

Three ways to decide who is poor

There are three recognised methods. They differ in one thing: how much of a household's income the government can see in its own records.

1

Means test — MT

Measure income directly. Add up income from every source, and test assets, using records the state already holds — tax, payroll, pensions, social security. Nothing is estimated.

Needs: most income visible in records, near-universal formal employment.

Used in: most OECD countries. Chile.

VERIFICATION

2

Proxy means test — PMT

Estimate income from proxies. Collect observable characteristics — housing, assets, household composition, education — and convert them into a score using weights estimated from a survey.

Needs: a household survey and an enumeration round.

Used in: Indonesia, Philippines, Pakistan, Albania, Malawi, Burkina Faso.

ESTIMATION

3

Hybrid means test — HMT

Both at once. Take the income that can be verified from records, estimate only the part that cannot — typically informal earnings — and add them together into one score.

Needs: linked administrative data and a unique ID, plus formal income large enough to be worth verifying.

Used in: Turkey. Common in Eastern Europe.

BOTH

The World Bank does not rank these as better or worse. It presents them as a menu, and says which one fits which situation: a hybrid test **“is recommended when formal income represents a large share of a household’s income”**, and it **“depends on the availability and quality of administrative data”**. That is the criterion that decides what a country can use.

FIGURE 8

How the decile is actually assigned

The survey records what can be observed. A model estimates the relationship between those observations and spending. That model is then applied to 93 million families, and the predicted values are ranked into deciles.

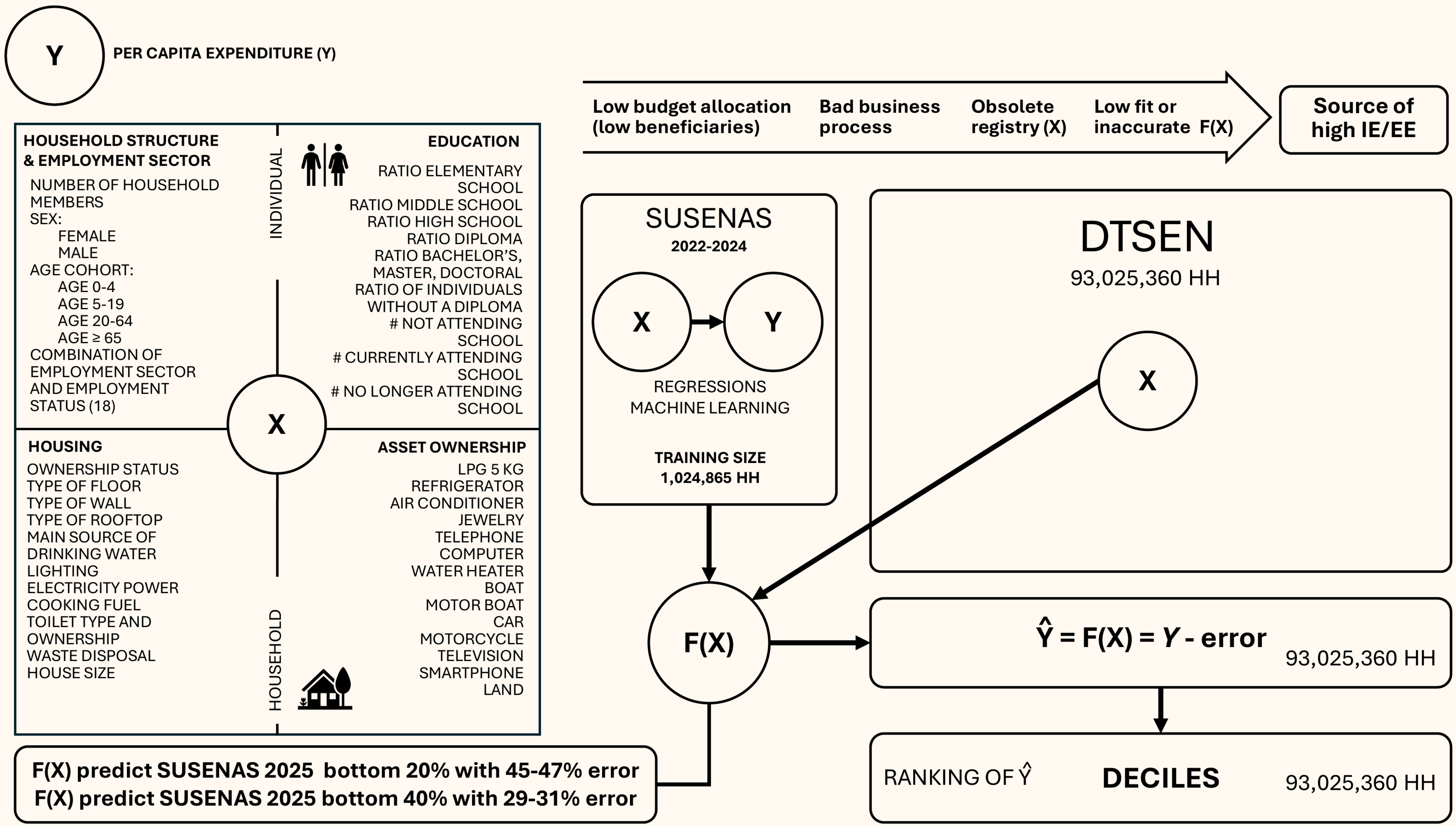


FIGURE 9

What the formula cannot see

The score is **$Y = X\beta + e$**. **X** is everything on the previous slide — housing, assets, schooling, household composition. **e** is everything else that decides whether a family is actually poor, and none of it is recorded anywhere.

Poorer than it looks — missed by the formula

Debt. A motorcycle and a refrigerator bought on credit count as assets. The instalments that consume the household's income do not.

Illness. Medical costs for a chronically sick member drain a family for years. The house and the assets look unchanged.

Lost work or a failed harvest. Income falls this season; the roof, the walls and the land are the same as last year.

Care at home. A disabled or elderly member needs a full-time carer, so one adult cannot work.

Supporting relatives elsewhere. Money sent to parents or children in another district leaves the household but never appears in the record.

Richer than it looks — wrongly included

Remittances. A son working in Jakarta or a daughter working abroad sends money home every month. The house still has a dirt floor.

A windfall. An inheritance, a land sale, a lottery win, compensation. Consumption rises; the recorded characteristics do not.

Savings. Cash and gold held at home are worth more than everything on the asset list, and are invisible.

Informal trade. A profitable warung or market stall is income without a payslip and without an asset that gets counted.

Support from adult children living separately but paying the bills.

Two consequences. First, **e runs in both directions**, which is why the same formula produces exclusion error and inclusion error at the same time. Second, **e is large**: the model predicts the bottom 20% with 45–47% error, and published work on the PKH formula puts its R^2 at 0.52 — about half of what determines household spending is outside X. That is also why Indonesia cannot simply copy Chile, which ranks households on recorded income: here, most income is informal and appears in no record.

FIGURE 10

Five sources of error. Only two are the formula.

Errors arise at two separate stages, and each stage produces both kinds of error. In the outcome data they look identical.

Choosing who to help — pensasaran

Delivering the help — penyaluran

1

Out-of-date register

The household's circumstances changed after it was last recorded.

TARGETING

2

Inaccurate formula

The proxies do not capture the household's real situation.

TARGETING

3

Budget limits

The list is cut to the funded quota, whatever the criteria say.

DELIVERY

4

Business process

Records enter unchecked, appeals go unanswered, payments are delayed.

DELIVERY

5

Elite capture

Local influence decides who appears on the list.

DELIVERY

IV What can digitalisation change?

Registration, verification and payment can be improved now. What the formula can know cannot.

FIGURE 11

Eight questions predict better than thirty-nine

The same Susenas data, five ways of building the score. The short model is not a compromise made for the sake of speed. It fits better in sample and predicts better out of sample — and it beats a machine-learning model using all 39 variables.

	SPECIFICATION	QUESTIONS	ADJ. R^2	POOREST DECILE IDENTIFIED	RICHEST DECILE IDENTIFIED	ERROR AT BOTTOM 20%	ERROR AT BOTTOM 40%
Model 1	Baseline, no fixed effects	**39**	0.511	38.2%	58.6%	48.6%	30.9%
Model 3	Baseline + district effects	**39**	0.575	40.9%	60.5%	45.8%	28.8%
XGBoost	Machine learning, district level	**39**	—	42.7%	60.6%	46.0%	29.7%
Model 4	**Parsimonious, no fixed effects**	8	**0.594**	**50.8%**	**63.2%**	**41.8%**	**27.8%**
Model 5	**Parsimonious + district effects**	8	**0.661**	**52.1%**	**65.6%**	**39.7%**	**26.0%**

This is what makes self-registration possible. A family can answer **eight** questions at a counter or on a phone, and answer them again when circumstances change. It cannot sit through a thirty-nine question interview, and neither can an agent doing several households a day. The short test is what a digital front door can actually carry — and here it is also the more accurate one.

FIGURE 12

What makes open registration work

Letting families come forward reaches poorer households than enrolling them automatically. In every country where it improved targeting, four things were put in place alongside it.

1

Staff to help people apply

19 of 20 on-demand social registries receive applications at staffed local offices. In Surabaya, **84%** of registrations came through an agent and only 16% were self-registered.

IN PLACE

2

Going out to find people

Brazil ran open registration for five years and reached 22 million families, and still found **over one million** more extremely poor families in a single year once it went looking for them.

IN PILOTS

3

Appeals that get answered

An appeal channel corrects the households the formula missed — but only if the appeals are actually decided.

PARTLY IN PLACE

4

Checking records before accepting them

Administrative data checked at the moment of application, before the record enters the register.

NOT YET AT INTAKE

Opening registration does help. A randomised trial in 400 Indonesian villages found that when households had to come forward and apply, the people who received benefits were **18 to 19% poorer** than under automatic enrolment, and the very poorest were twice as likely to be reached. But only about **60%** of the very poorest applied — and making the process harder screened out poor and rich equally, so any extra difficulty is a cost borne by the poor for no gain.

FIGURE 13

How digitalisation changes targeting accuracy

Each part of the digital infrastructure acts on a different step, and moves a different error. The last column is the one that matters for planning: where each one stops.

WHAT IT DOES	WHAT IT CHANGES	EXCLUSION	INCLUSION	WHERE IT STOPS
Digital identity	One verified identity per person. Duplicates, deceased and ghost records removed.	↓	↓↓	Cannot see anyone who has never registered.
Data exchange at the moment of application	Land, vehicles, formal employment and electricity use checked *before* the record is accepted, not after.	—	↓↓↓	Screens out the clearly non-poor. It does not measure income, so it cannot rank the informal majority.
Open registration	A family that was missed can apply at any time instead of waiting for the next survey round.	↓↓↓	↑	Only about 60% of the poorest come forward on their own. Outreach and assistance close the rest.
Grievance that is decided	Wrong decisions corrected in both directions, and the pattern of appeals shows where the formula is failing.	↓↓	↓↓	An upheld appeal needs a funded place to move into.
Continuous re-ranking	The register keeps up as circumstances change, instead of ageing between survey rounds.	↓↓	↓↓	Still a ranking of proxies, and still only as fresh as the last rebuild.
Digital payment	Money reaches the account. New recipients are onboarded without a months-long wait for an account to be opened.	↓↓	—	Fixes delivery, not selection. It cannot correct a wrong list.

Two things none of this touches. It does not create **places** — Figure 4 — and it does not make income **observable**, so the error in the formula itself stays until more of the economy is formal. Everything digitalisation does well sits between those two limits.

CONCLUSION

What digitalisation can and cannot fix

Of the five sources of error, the two that digitalisation closes fastest are not the statistical formula.

SOURCE OF ERROR	CAN DIGITALISATION FIX IT?	
Out-of-date register	**Yes, directly**	Continuous registration and administrative updates keep the record current.
Business process	**Yes, and fastest**	Check records at the point of application; decide appeals; pay without delay.
Elite capture	**Partly**	An audit trail and published lists make local discretion visible.
Inaccurate formula	**Only up to a limit**	Bounded by how much income can be verified. It improves as formal employment grows, not as the model improves.
Budget limits	**No**	Not a data problem. Better data changes who receives, not how many.

The digital public infrastructure described earlier today is necessary. This is a map of what it will and will not fix.

Appendix

Reference table, for questions.

APPENDIX

Every figure with the population it is measured against

Exclusion and inclusion error can only be compared when you know what each was measured against. These were not measured the same way. The sixth column says how each one differs.

COUNTRY	PROGRAMME	YEAR	EXCLUSION	INCLUSION	MEASURED AGAINST	SOURCE
Turkey	ESSN (refugee cash transfer)	2019	34%	24%	Applicants only — 1.6m of about 3.2m refugees, surveyed before any payment was made. Not comparable with the rest of the table.	Cuevas, Inan, Twose & Çelik (2019), World Bank and WFP
Turkey	Green Card	2003	88%	45%	Poorest decile.	World Bank (2013), Table 3
Turkey	Green Card	2008	38%	29%	Poorest decile. Same programme, five years later.	World Bank (2013), Table 3
Georgia	TSA	2013	54%	—	Bottom quintile of consumption, measured after transfers are removed.	World Bank (2020)
Georgia	TSA	2018	64%	—	As above. About a third of the increase came from prices being updated while the coefficients were not.	World Bank (2020)
Egypt	Takaful	2017	~85%	33%	Exclusion is derived from reported coverage, not stated by IFPRI. Inclusion is measured among beneficiaries, and the population is households with children.	Breisinger et al. (2018), IFPRI
Tunisia	PNAFN	2015	83%	60%	Exclusion against the poorest **decile**; inclusion against the poorest **quintile**. The two are not on the same base.	World Bank (2022)
Tunisia	PNAFN and AMG2	2015	52%	57%	Both against the poorest quintile.	World Bank (2022)
Morocco	RAMED	2019–20	~50%	70%	Exclusion covers RAMED and Tayssir together, against the poorest quintile. Inclusion is the share of beneficiaries outside that quintile.	World Bank (2021)
Jordan	NAF	2010	69%	—	Bottom decile. A second World Bank source reports 83.5% against the poorest quintile — a different denominator, not a contradiction.	World Bank (2012); Silva, Levin & Morgandi (2012)
Chile	Solidario	2002	—	38.6%	Entrants outside the poorest ventile.	Larrañaga & Contreras (2010)
Chile	Solidario	2008	—	1.0%	As above, after local cut-offs were replaced by a single national cut-off. The formula itself did not change.	Larrañaga & Contreras (2010)
Indonesia	PKH	2025	76.6%	46.0%	Deciles 1–4 with programme conditions, households.	Susenas March 2025, own calculation
Indonesia	BPNT	2025	74.4%	36.1%	Deciles 1–5, households.	Susenas March 2025, own calculation
Indonesia	PBI	2025	42.6%	40.9%	Deciles 1–5, individuals.	Susenas March 2025, own calculation
Indonesia	PIP	2025	82.1%	44.2%	Deciles 1–4, school age 6–27, individuals.	Susenas March 2025, own calculation

Sources. Susenas March 2023, 2024 and 2025, own calculation following the definitions of the DEN Indonesia Social Protection Targeting Monitor. Cross-country figures from World Bank and IFPRI publications, each with its own definition of the target population — see the note under each figure. Framework and illustrations from *Salah Sasaran* (Mei 2026). Audit findings from BPK RI, IHPS II 2024.